



Petar Jolakoski

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<https://scholar.google.com/citations?user=CWn3tBgAAAAJ&hl=en> | **Website:**

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EDUCATION AND TRAINING

01/02/2023 – CURRENT Skopje, North Macedonia

PHD STUDENT IN COMPUTER SCIENCE AND ENGINEERING Faculty of Computer Science and Engineering

- Scientific and research ethics
- Methodology of scientific research in computer sciences and engineering
- Introduction to dynamical systems and control
- Analysis and evolution of complex networks
- Advanced topics in data science
- Theory of stochastic processes

Website <https://www.finki.ukim.mk/en> |

Thesis "Stochastic resetting: A mechanism for optimization and control in physics, economics and machine learning"

31/10/2019 – 19/01/2021 Skopje, North Macedonia

MSC IN ECONOMICS (STATISTICAL METHODS) Faculty of Economics - Skopje

- Theoretical econometrics;
- Multivariate analysis;
- Mathematical statistics;
- Time series analysis;
- Research methods;
- Advanced econometrics;
- Financial econometrics.

Website <https://eccf.ukim.edu.mk/en/home-default/> |

Thesis Thesis: "Socio-economic determinants of the Covid-19 pandemic: A Bayesian approach"

14/09/2015 – 30/06/2019 Skopje, North Macedonia

BACHELOR OF ECONOMICS (ACCOUNTING AND AUDITING) University of "Ss. Cyril and Methodius", Faculty of Economics

Website <http://www.eccf.ukim.edu.mk/> |

Thesis "Value creation and value capture in M&A markets: A cooperative game theory approach"

30/06/2018 – 21/07/2018 Milan, Italy

COURSE CREDITS Bocconi University

Certificate for participation and successful completion of the exam in the Bocconi Summer School 2018

Name of the course: Equity and Venture Capital Financing

Subject group: Economics of Financial Intermediaries

Address Via Roberto Sarfatti, 25, Milan, Italy | **Website** <https://www.unibocconi.eu>

● WORK EXPERIENCE

JUNIOR RESEARCH ASSOCIATE – MACEDONIAN ACADEMY OF SCIENCES AND ARTS – 30/11/2021 – Current – SKOPJE, NORTH MACEDONIA

Department: Research Center for Computer Science and Information Technologies | **Email:** jolakoskip@manu.edu.mk

Conducting research and writing research project manuscripts in:

- **Mathematical sciences:** Stochastic Processes;
- **Physics:** Statistical Physics, Nonlinear Dynamics;
- **Finance:** Option pricing;
- **Economics:** Economic inequality and mobility;

LECTURER – BRAINSTER NEXT COLLEGE – 01/08/2022 – 28/02/2025 – SKOPJE, NORTH MACEDONIA

Complete lectures, exercises and lab exercises in Jupyter notebooks (~10 hours per week) in the winter semester for the following subjects:

- **Linear algebra** (Vectors, Matrices, Linear transformations, Systems of linear equations, Vector spaces and subspaces, Determinants, Eigenvalues and eigenvectors, Singular Value Decomposition).
- **Probability and statistics** (Descriptive statistics, Probability theory, Inferential statistics: hypothesis testing, ANOVA, sampling distributions, Single and multi-variable linear regression, Logistic regression).

ASSISTANT LECTURER – BRAINSTER – 31/08/2020 – 01/08/2023 – SKOPJE, NORTH MACEDONIA

Lecture and workshops (10 hours per week) in the following areas:

- Linear algebra;
- Single variable calculus;
- Multivariable calculus;
- Descriptive statistics;
- Intro to probability and probability distributions;
- Parameter estimation (MLE and Bayesian).

RESEARCH ASSISTANT – ZMAI – 31/12/2018 – 01/04/2022 – SKOPJE, NORTH MACEDONIA

- Professional experience in data analysis;
- Conducting research and writing research project manuscripts;
- Most recent research: The short-term effect of the coronavirus pandemic on income distribution in the Republic of Northern Macedonia.

UNDERGRADUATE TEACHING ASSISTANT – FACULTY OF ECONOMICS - SKOPJE – 31/08/2019 – 14/01/2021 – SKOPJE, NORTH MACEDONIA

Conducting exercises in introductory classes in Mathematics and Statistics on a weekly basis. The classes include:

- **Mathematics:** Sequences, Limits, Linear Algebra, Calculus, Long-Term Financial Operations;
- **Statistics:** Descriptive analysis, Random variables and probability distributions, Samples, Inference, Hypothesis testing, ANOVA, Chi-square test, Simple linear regression and correlation, Multiple linear regression and correlation, Indices, Time series analysis;
- **Introductory Econometrics and Financial Econometrics:** Mathematical and statistical foundations for Econometrics, Linear regression model (review), Univariate time series modelling and forecasting, Multivariate models, Long-run relationships in finance, Stochastic modelling of volatility and correlation;
- **Statistical quality control:** Basic analysis of control charts for numerical variables and attributes, basic process capability analysis.

ACCOUNTANT ANALYST – JOLAK DOO – 31/05/2016 – 31/08/2016 – OHRID, NORTH MACEDONIA

- evaluating financial statements;

- ensuring that the financial statements are in compliance with IFRS.

● JOB-RELATED SKILLS

Job-related skills

- **Highly adaptable, rapid learner:** over time I shifted from accounting degree to physics- and ML-driven research; comfortable in identifying knowledge gaps early and closing them quickly through focus and implementation.
- **Self-directed researcher with strong independence and autonomy:** independently designed and executed research projects end-to-end from problem formulation and coordinating team responsibilities to experiment design and implementation, evaluation and manuscript preparation.
- **Intellectually curious with strong problem-solving abilities:** breaking down ambiguous technical problems, testing hypotheses systematically, and communicate results clearly with collaborators across different disciplines, from business-oriented stakeholders and social scientists to physicists.
- **Statistical computing and data analysis:** Python, R and MATLAB for simulation, modeling and inference; building reproducible pipelines for data preparation, evaluation and visualization.

● SKILLS

Python for Scientific Computing | Linear algebra | Probability & Statistics | Calculus | Machine learning | Claude Code | MATLABSimulink | R for statistics | Microsoft Office | Wolfram Mathematica (Basic) | SQL (basics) | Learning theory | Pedagogy

● PUBLICATIONS

2025

[Trajanovski, P., Jolakoski, P., Kocarev, L., Metzler, R. and Sandev, T., 2025. Generalised Ornstein-Uhlenbeck process: memory effects and resetting. Journal of Physics A: Mathematical and Theoretical, 58\(4\), p.045001.](#)

Journal of Physics A: Mathematical and Theoretical

2025

[Jolakoski, P., Trajanovski, P., Iomin, A., Kocarev, L. and Sandev, T., 2025. Mean first-passage time of heterogeneous telegrapher's process under stochastic resetting. Europhysics Letters, 149\(4\), p.41004.](#)

Europhysics Letters

2025

[Jolakoski, P., Trajanovski, P., Pal, A., Stojkoski, V., Kocarev, L. and Sandev, T., 2025. Impact of stochastic resetting on resource allocation: The case of reallocating geometric Brownian motion. Physical Review E, 111\(3\), p.034129.](#)

Physical Review E

2025

[Jolakoski, P. et al. \(2025\) 'Response to an external field of a generalized Langevin equation with stochastic resetting of the memory kernel', Phys. Rev. E, 112, p. 024120.](#)

Physical Review E

2025

[Pesovski, I., Jolakoski, P., Trajkovik, V., Kubincova, Z. and Herzog, M.A., 2025. Predicting Student Achievement through Peer Network Analysis for Timely Personalization via Generative AI. Computers and Education: Artificial Intelligence, p.100430.](#)

Computers and Education: Artificial Intelligence

2023

[Jolakoski, P., Pal, A., Sandev, T., Kocarev, L., Metzler, R. and Stojkoski, V., 2023. A first passage under resetting approach to income dynamics. Chaos, Solitons & Fractals, 175, p.113921.](#)

Chaos, Solitons & Fractals

2023

[Disoska, E.M., Toshevsk-Trpchevsk, K., Tevdovski, D., Jolakoski, P. and Stojkoski, V., 2023. A Pooled Overview of the European National Innovation Systems Through the Lenses of the Community Innovation Survey. Journal of the Knowledge Economy, pp.1-25.](#)

Journal of the Knowledge Economy

2023

[Trajanovski, P., Jolakoski, P., Kocarev, L. and Sandev, T., 2023. Ornstein-Uhlenbeck Process on Three-Dimensional Comb under Stochastic Resetting. Mathematics, 11\(16\), p.3576.](#)

Mathematics

2023

[Trajanovski, P., Jolakoski, P., Zelenkovski, K., Iomin, A., Kocarev, L. and Sandev, T., 2023. Ornstein-Uhlenbeck process and generalizations: Particle dynamics under comb constraints and stochastic resetting. Physical Review E, 107\(5\), p.054129.](#)

Physical Review E

2022

[Stojkoski, V., Utkovski, Z., Jolakoski, P., Tevdovski, D. and Kocarev, L., 2022. Correlates of the country differences in the infection and mortality rates during the first wave of the COVID-19 pandemic: evidence from Bayesian model averaging. Scientific reports, 12\(1\), pp.1-11.](#)

2022

[Tevdovski, D., Madjoska, J., Jolakoski, P., Jovanovic, B. and Stojkoski, V., 2022. Firm profits and government activity: An empirical investigation. Croatian Economic Survey, 24\(1\), pp.43-82.](#)

Croatian Economic Survey

2022

[Tevdovski, D., Jolakoski, P. and Stojkoski, V., 2022. Determinants of budget deficits: The effects of the COVID-19 crisis. Economic Annals, 67\(232\), pp.105-126.](#)

Economic Annals

2022

[Tevdovski, D., Jolakoski, P. and Stojkoski, V., 2022. The impact of state capacity on the cross-country variations in COVID-19 vaccination rates. International Journal of Health Economics and Management, 22\(3\), pp.237-255.](#)

International Journal of Health Economics and Management, pp.1-19.

2021

[Stojkoski, V., Jolakoski, P. and Ivanovski, I., 2021. The short-run impact of COVID-19 on the activity in the insurance industry in the Republic of North Macedonia. Risk Management and Insurance Review, 24\(3\), pp.221-242.](#)

Risk Management and Insurance Review

● **ONGOING WORK**

Physics-inspired regularization in ML

During 2025 I started exploring physics-inspired regularization for machine learning analysis of anomalous diffusion. This paper is currently under review and will be the final chapter of my PhD thesis: "Stochastic resetting: A mechanism for optimization and control in physics, economics and machine learning"

Applying wavelet analysis to business cycle synchronization

Currently I am working on a research paper implementing advanced time-series analysis technique based on wavelet analysis to better understand business cycle synchronization in time-frequency space.

● **HOBBIES AND INTERESTS**

Playing drums

I've been playing drums for approximately a year. One of the best decisions ever. I practice ~1 hour per day on average.

Learning Physics and Mathematics

Ongoing obsession for approximately 6 years. Currently strenghtening my fundamentals in:

- Classical mechanics
- Trigonometry
- Calculus
- Linear algebra
- Probability & Statistics
- Scientific ML

I spend ~1.5 hours per day on average studying topics from these subjects. In general, I apply learning theory to better commit everything relevant to long-term memory, mainly through spaced repetition techniques and knowledge graphs.

Various sports

Football, Tennis, Running.

Film

Curiosity about film as a narrative and visual art.

● **LANGUAGE SKILLS**

Mother tongue(s): **MACEDONIAN**
Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
ITALIAN	A1	A1	A1	A1	A1
GERMAN	A1	A1	A1	A1	A1

● CONFERENCES AND SEMINARS

26/03/2023 – 31/03/2023 Dresden, Germany

Deutsche Physikalische Gesellschaft (DPG) - Spring Meeting (SKM23)

Poster presentation: "Ornstein-Uhlenbeck process and generalizations: influence of comb geometry and stochastic resetting on the particle dynamics"

17/03/2024 – 22/03/2024 Berlin, Germany

Deutsche Physikalische Gesellschaft (DPG) - Spring Meeting (SKM24)

Poster presentation: "Disaggregating temporal properties of individual income dynamics using first-passage under resetting approach"

15/09/2022 – 18/09/2022 Ohrid, Macedonia

14th Conference of the Society of Physicists of Macedonia - CSPM 2022

Poster presentation: "Non-ergodicity and first-passage dynamics in economics"

19/09/2024 – 22/09/2024 Ohrid, Macedonia

15th Conference of the Society of Physicists of Macedonia - CSPM 2024

Poster presentation: "The effect of resetting in the non-ergodic regime of reallocating geometric brownian motion"

● HONOURS AND AWARDS

Honours and awards

Honorable mentions: I was recognized as one of the best students for 2016/17 and 2017/18 with GPA 10.00/10.00 for the relevant student years.

Honours and awards

Second place at the International Mathematics Competition KENGUR held on March 20, 2014

● CERTIFICATIONS

Certifications

Certificate of Achievement: Together with my Team, I was recognized as a winner and one of the best individuals in a business case competition, part of EY Program for Development of Young Potentials 2018, organized in cooperation with the Faculty of Economics and Faculty of Law at the University of Ss. Cyril and Methodius in Skopje.

Certifications

CFA Institute Research Challenge hosted by CFA Slovenia

The CFA Institute Research Challenge is a global competition which tests the analytic, valuation, reportwriting, and presentation skills of university students. In this competition, I worked with a team of students to prepare a comprehensive sell-side initiating coverage report on Sava Re Group. Our team then took our findings before a panel of experts from financial institutions in Switzerland, Slovenia, Croatia and defended our recommendation. We achieved a second place in the Local Finals organized by CFA Society Slovenia.